

Assessment of Alzheimer's Disease Progression Using Comorbidity-Aware Meta-Bayesian Framework with Resting-State fMRI

Second Review

Harishkanna R (3122235001302)
Dhananchezhiyan M (3122235001041)
Nagindhar Sachein R L (3122235001702)

Guided by: Dr. N. Ahana Priyanka, Assistant Professor

Department of Computer Science and Engineering
Sri Sivasubramaniya Nadar College of Engineering

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First Review - Queries and Answers

Panel comment	Response	Detail in
Need more clarity on the dataset (biomarkers)	175 audited, 165 preprocessed, 127 subjects. Four biomarkers on 246 regions.	Results slides; report §6.1
Need more clarity on the Meta-Bayesian framework	“Meta”: biomarkers as tasks. “Bayesian”: ordinal head with uncertainty. Design only.	Core Algorithm (design); report §4.5
Need more clarity on the comorbidity data	Seven covariates with source and status; clinical tables cleaned for all six stages.	Modules slide; report §4.5

Problem Statement

Problem

Existing systems analyse rs-fMRI biomarkers **independently** and omit clinical context, giving **overconfident, deterministic staging errors** on borderline subjects.

- ▶ **Context:** dementia progresses through six ordered stages.
- ▶ **Limitation:** pipelines isolate ALFF, ReHo, DC and FC.
- ▶ **Clinical gap:** models ignore how comorbidities modify neurodegeneration.

Research question

Can a common AD progression representation be found across several functional biomarkers, accounting for comorbidity and prediction uncertainty?

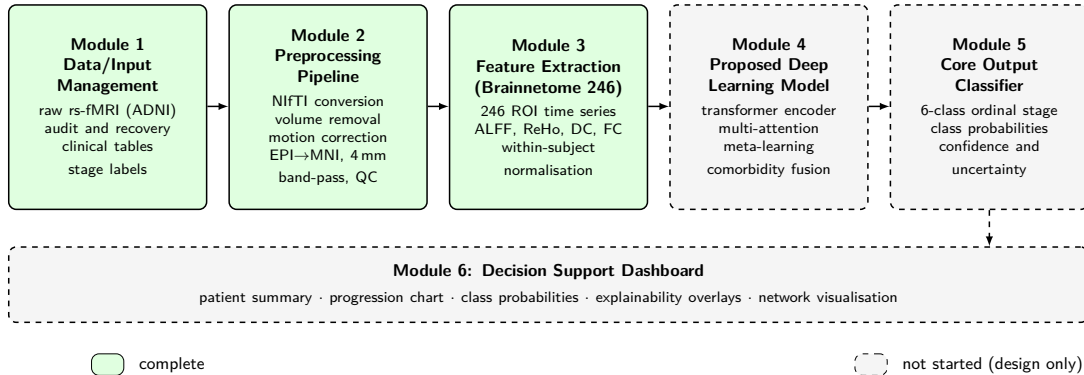
Literature update: 2026 papers cover ordinal staging and uncertainty; none of the three checked uses cross-biomarker meta-learning (report §1.3).

Project Objectives

#	Review I objective	Status
1	Identify functional biomarkers and comorbidity patterns	Partly
2	Develop a comorbidity-aware feature representation	Partly
3	Design a Meta-Bayesian learning framework	Not started
4	Incorporate uncertainty estimation	Not started
5	Validate on benchmark data: performance, statistics, clinical	Not started

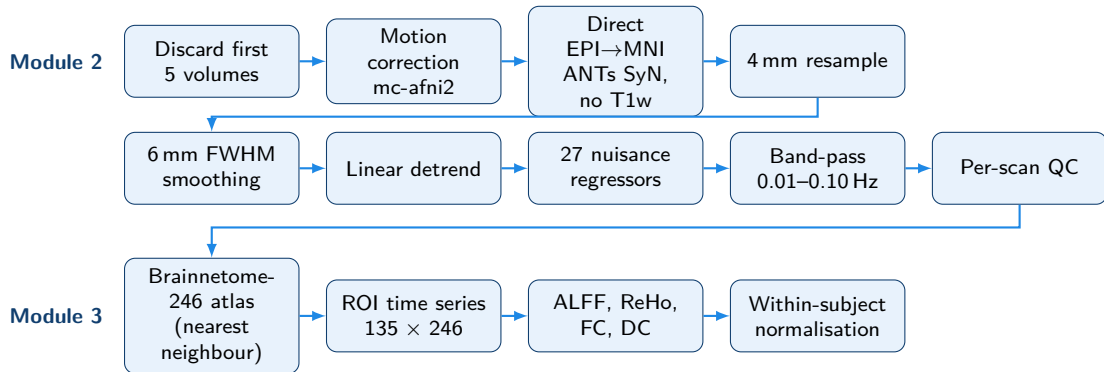
- ▶ Objectives 1–2: the **data inputs** (imaging biomarkers and cleaned clinical tables) are ready.
- ▶ Comorbidity pattern analysis and fusion belong to the model stage, not yet started.

Proposed System Architecture



System Modules

Module (Review I names)	Status	At Review II
1 Data/Input Management	Complete	Imaging and clinical tables complete and cleaned
2 Preprocessing Pipeline	Complete	165 of 173 scans; 8 rejected (TR 6.02 s)
3 Feature Extraction (Brainnetome 246)	Complete	ALFF, ReHo, DC, FC for 165/165; normalised
4 Proposed Deep Learning Model	Not started	Transformer, meta-learning, fusion: design only
5 Core Output Classifier	Not started	Bayesian ordinal head: design only
6 Decision Support Dashboard	Not started	No interface built



Every step is checked per acquisition; a scan that fails a check is logged, never silently altered.

Core Algorithm A: Implemented Biomarkers and Normalisation

Biomarkers (per Brainnetome region)

- ▶ **ALFF**: mean FFT amplitude, 0.01–0.10 Hz
- ▶ **ReHo**: Kendall's W , 27-voxel neighbourhood
- ▶ **FC**: 246×246 Pearson, signed, unthresholded
- ▶ **DC**: $DC_i = \sum_{j \neq i} FC_{ij}$

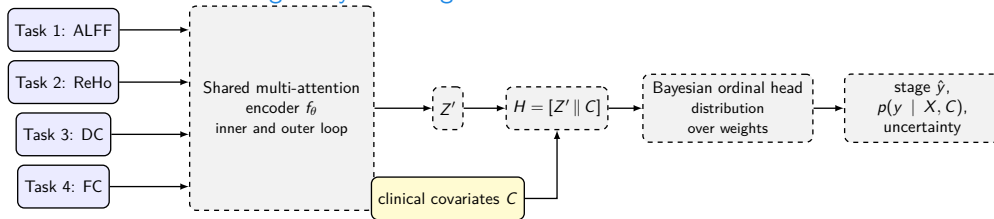
Within-subject normalisation

- ▶ $mALFF = ALFF_i / \overline{ALFF}$
- ▶ $mReHo = ReHo_i / \overline{ReHo}$
- ▶ $DC_z = (DC_i - \overline{DC}) / \sigma_{DC}$
- ▶ $FC \rightarrow \text{artanh}(\text{clip}(r, \pm 0.999999))$, diagonal = 0

- ▶ Every statistic comes from **one acquisition's own 246 values**.
- ▶ No cross-subject statistic and no label: leakage-free by construction.
- ▶ Verified: $\text{mean}(mALFF) = \text{mean}(mReHo) = 1.000000$; $\text{std}(DC_z) = 1.000000$ (165/165).
- ▶ 0 off-diagonal entries needed clipping; Fisher-z diagonal exactly 0.

Core Algorithm B: Meta-Bayesian Pipeline (Design, Not Implemented)

Design only: nothing is trained and no result exists.



MAML update over biomarkers

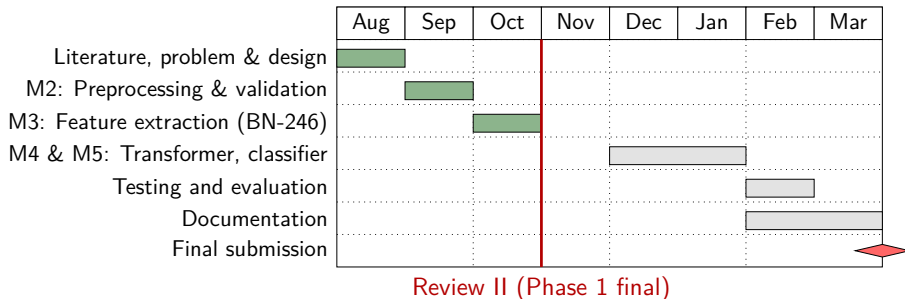
$$\theta'_b = \theta - \alpha \nabla_\theta \mathcal{L}_b^{\text{sup}}(\theta)$$
$$\theta \leftarrow \theta - \beta \nabla_\theta \sum_b \mathcal{L}_b^{\text{qry}}(\theta'_b)$$

Ordinal head (cumulative link)

$$P(y \leq k \mid H, w) = \sigma(\tau_k - g_w(H)), \quad \tau_0 < \dots < \tau_4$$

Support and query sets are subject-disjoint, within training folds. Uncertainty = predictive entropy over S weight samples.

Implementation Status



One milestone ahead of the Review I plan

Module 2 was planned for Review II and Module 3 for Review III. **Both are complete at Review II**, the final review of Phase 1.

Applies to the **data foundation only** (Modules 1–3); Modules 4–6 are not started. Module 2 completed in September; Module 3 in October (normalised biomarkers 5 Oct).

Pipeline Outputs

```
derivatives/  
  fsfast/<STAGE>/sub-*/ses-*/run-*/  
    *_desc-preproc_bold.nii.gz + .json  
  qc/          QC metrics and plots  
  biomarkers/   23 files  
  biomarkers_normalized/<STAGE>/.../  
                    10 files  
  fsfast/FINAL_6GROUP_AUDIT/  
  fsfast/BIOMARKER_DATASET_AUDIT/  
  STAGE_HEATMAPS/
```

- ▶ 165 acquisitions, same layout each.
- ▶ Each carries a **provenance record**: parameters, versions, source checksum.
- ▶ Resume re-validates files on disk, not the checkpoint.
- ▶ Raw biomarkers are never modified.

No interface exists

These are data outputs. The Module 6 dashboard is **not built**.

Counts are per acquisition; the one pilot acquisition keeps its original extra files.

Results: Data Foundation Verification

These are **data-stage** results. **No model performance exists.**

Stage	Elig.	Prep.	Bio.	Subj.
CN	29	27	27	26
SMC	32	31	31	21
EMCI	30	29	29	26
MCI	35	32	32	14
LMCI	22	21	21	20
AD	25	25	25	20
Total	173	165	165	127

Check (165 acquisitions)	Result
246/246 regions covered	165/165
ROI series (135, 246)	165/165
NaN / Inf / zero-variance	0 / 0 / 0
Pilot reproduction	bit-identical
DC recomputed from FC, max diff.	0.000e+00
Consistency checks passed	165/165

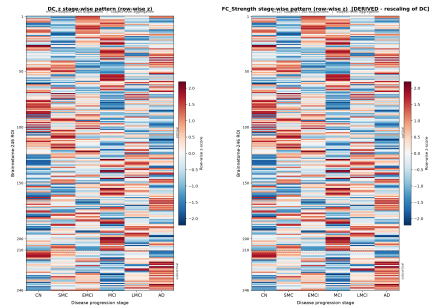
- ▶ 127 subjects, not 165 scans: 96 have one scan, 25 two, 5 three, 1 four.
- ▶ Smallest class: MCI, 32 scans but only **14 subjects**.
- ▶ QC medians ($n = 164$): tSNR 127.95, SNR 13.12, FD 0.317 mm.

Finding 1: FC Strength Equals DC/245

Review I fourth column was redundant

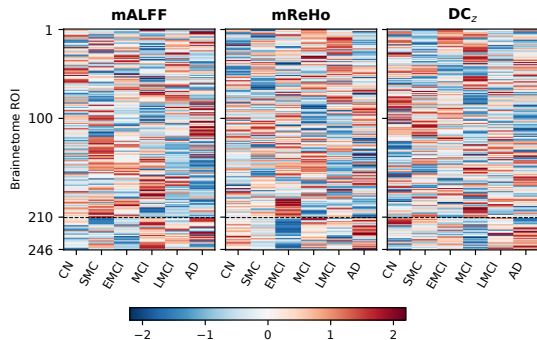
$$FCS_i = \frac{1}{245} \sum_{j \neq i} FC_{ij} = \frac{DC_i}{245}$$

- ▶ Max difference **6.939e-18**; Pearson $r = 1.000000000000$.
- ▶ **All four biomarkers are kept.**
- ▶ ALFF, ReHo, DC: 246×3 regional matrix.
- ▶ FC: full 246×246 Fisher-z matrix.
- ▶ A design correction found by auditing our own features.



DC_z (left) and FC_Strength (right): identical panels, as expected. Descriptive and exploratory only.

Finding 2: No Univariate Progression Signal



Stage-wise row-z patterns, 127 subjects (CN→AD).

Descriptive and exploratory; no statistical test.

	Monotonic CN→AD	By chance
mALFF	1 / 246	≈ 0.7
mReHo	0 / 246	≈ 0.7
DC _z	1 / 246	≈ 0.7

- ▶ No more than chance; stage variation is 4–5× below anatomy.
- ▶ Informative, not a failure.
- ▶ Group means discard the whole connectivity structure.
- ▶ **Motivates the multivariate model.**

Team Responsibilities

Team member	Responsibility	Status at Review II
Harishkanna R	Preprocessing (Module 2); transformer model (Module 4)	Preprocessing: Complete Transformer: Not started
Dhananchezhian M	Meta-Bayesian module (meta-learning, Bayesian classifier) and final module (dashboard)	Not started
Nagindhar Sachein R L	Dataset cleaning (clinical and comorbidity tables, dataset expansion)	Clinical tables cleaned; dataset expansion next

Feature extraction (Module 3) was completed by the team and is verified.

Next Steps

- ① Build subject-grouped cross-validation folds.
- ② Expand the dataset by **+50 acquisitions per stage**.
- ③ Then implement Modules 4–5 (Dec–Jan in the Review I plan).

Where we stand

The data foundation is complete and independently verified: a **full milestone ahead of plan**. The modelling stages are next; no model result exists yet.

The expansion helps the 127-subject limit only through *new subjects*; it has not started.

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Thank You

Questions and Discussion